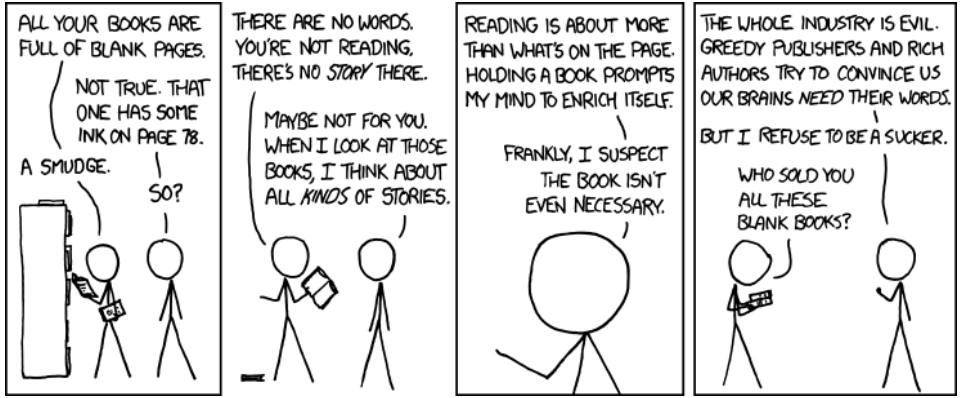
xAI-Proj-B
Bachelor Project
Explainable Machine Learning

Sebastian Dörrich (MSc)
Chair of Explainable Machine Learning (xAI)

Resources

Literature

- Google: [Link]
- Google Scholar: [Link]
- Bamberg Catalogue: [Link]
- Springer Link: [Link]
- PubMed: [Link]
- ResearchGate: [Link]
- ArXiv: [Link]
- Research Rabbit: [Link]
- Connected Papers: [Link]



Munroe, R. Xkcd. Alternative Literature. https://xkcd.com/971/

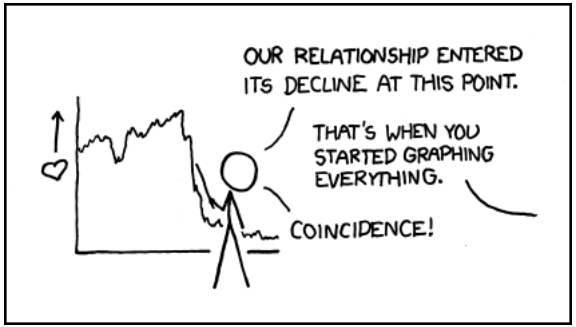
Images and Graphs

Images

- Pexels: [Link]
- Unsplash: [Link]
- Adobe Express: [Link]
- Canva: [Link]
- PngTree: [Link]

Graphs

- Draw.io: [Link]
- Lucid: [Link]

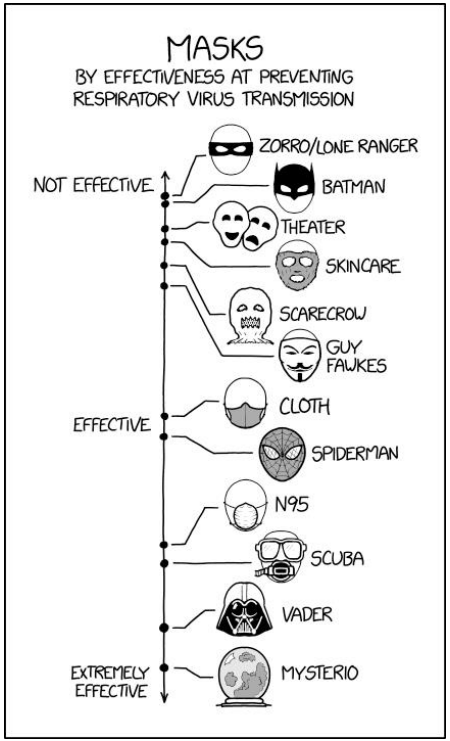


Munroe, R. Xkcd. Decline. https://xkcd.com/523/

Image Enhancer

Be advised that for all these tools, you upload your images to the internet and will most likely revoke your rights for the images by doing so !!

- Remove Image Background: [Link]
- Upscale.media: [Link]
- Magic Eraser: [Link]
- Animated Gif Editor: [Link]
- BeFunky: [Link]
- LunaPic: [Link]



Munroe, R. Xkcd. Masks. https://xkcd.com/2367/

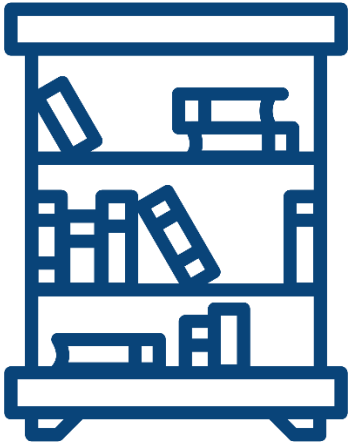
Two Essentials

- Gimp: [Link]
  - Tutorials: https://www.gimp.org/tutorials/

- Inkscape: [Link]
  - Extract Images from PDFs: https://juanjose.garciaripoll.com/blog/using-inkscape-with-pdf-files/index.html

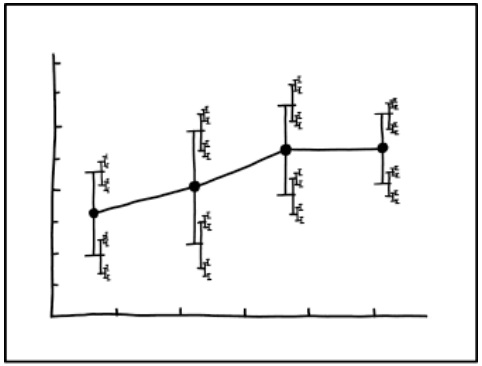
Your Tools

Reference Management



Reference Managers

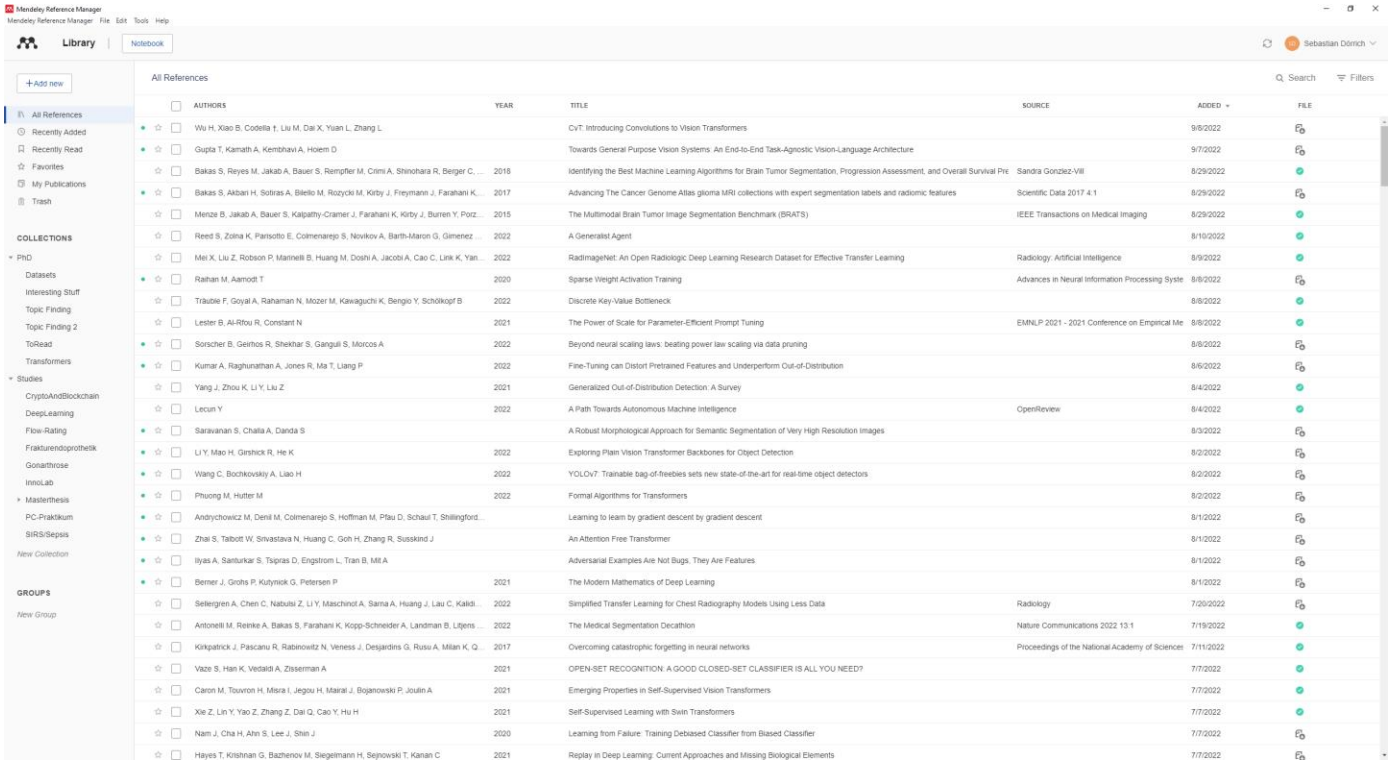
- Facilitate your citation work
- Create your own literature database (“library”)
- Are only as good as the user -> error prone!



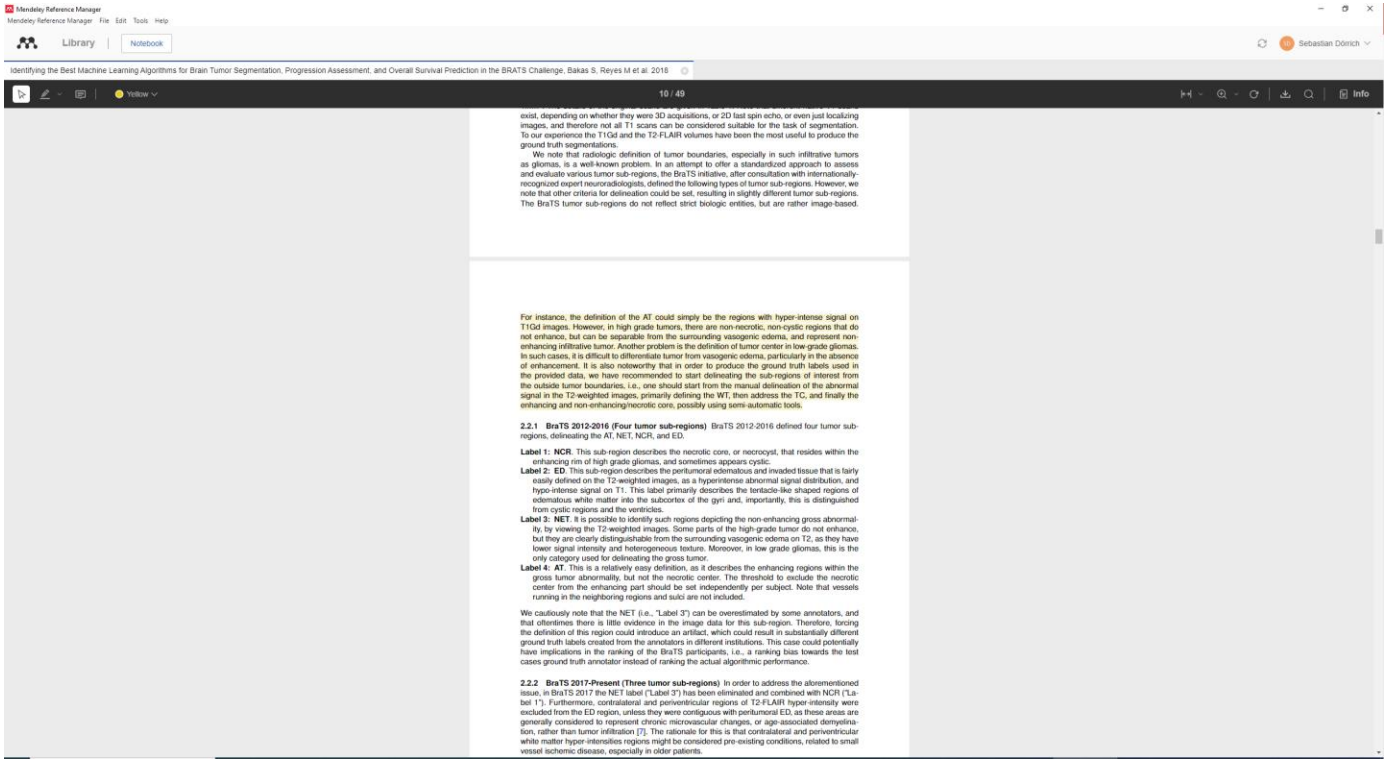
I DON'T KNOW HOW TO PROPAGATE ERROR CORRECTLY, SO I JUST PUT ERROR BARS ON ALL MY ERROR BARS.

Munroe, R. Xkcd. Error Bars. https://xkcd.com/2110/

Mendeley Reference Manager



Mendeley Reference Manager – Viewer



exist, depending on whether they were 3D acquisitions, or 2D fast spin echo, or even just localizing images, and therefore not all T1 scans can be considered suitable for the task of segmentation. To our experience the T1Gd and the T2-FLAIR volumes have been the most useful to produce the ground truth segmentations.

We note that radiologic definition of tumor boundaries, especially in such infiltrative tumors as gliomas, is a well-known problem. In an attempt to offer a standardized approach to assess and evaluate various tumor sub-regions, the BraTS initiative, after consultation with internationally-recognized expert neuroradiologists, defined the following types of tumor sub-regions. However, we note that other criteria for delineation could be set, resulting in slightly different tumor sub-regions. The BraTS tumor sub-regions do not reflect strict biologic entities, but are rather image-based.

For instance, the definition of the AT could simply be the regions with hyper-intense signal on T1Gd images. However, in high grade tumors, there are non-necrotic, non-cystic regions that do not enhance, but can be separable from the surrounding vasogenic edema, and represent non-enhancing infiltrative tumor. Another problem is the definition of tumor center in low-grade gliomas. In such cases, it is difficult to differentiate tumor from vasogenic edema, particularly in the absence of enhancement. It is also noteworthy that in order to produce the ground truth labels used in the provided data, we have recommended to start delineating the sub-regions of interest from the outside tumor boundaries, i.e., one should start from the manual delineation of the abnormal signal in the T2-weighted images, primarily defining the WT, then address the TC, and finally the enhancing and non-enhancing/necrotic core, possibly using semi-automatic tools.

2.2.1 BraTS 2012-2016 (Four tumor sub-regions) BraTS 2012-2016 defined four tumor sub-regions, delineating the AT, NET, NCR, and ED.

Label 1: NCR. This sub-region describes the necrotic core, or necrocyst, that resides within the enhancing rim of high grade gliomas, and sometimes appears cystic.

Label 2: ED. This sub-region describes the peritumoral edematous and invaded tissue that is fairly easily defined on the T2-weighted images, as a hyperintense abnormal signal distribution, and hypo-intense signal on T1. This label primarily describes the tentacle-like shaped regions of edematous white matter into the subcortex of the gyri and, importantly, this is distinguished from cystic regions and the ventricles.

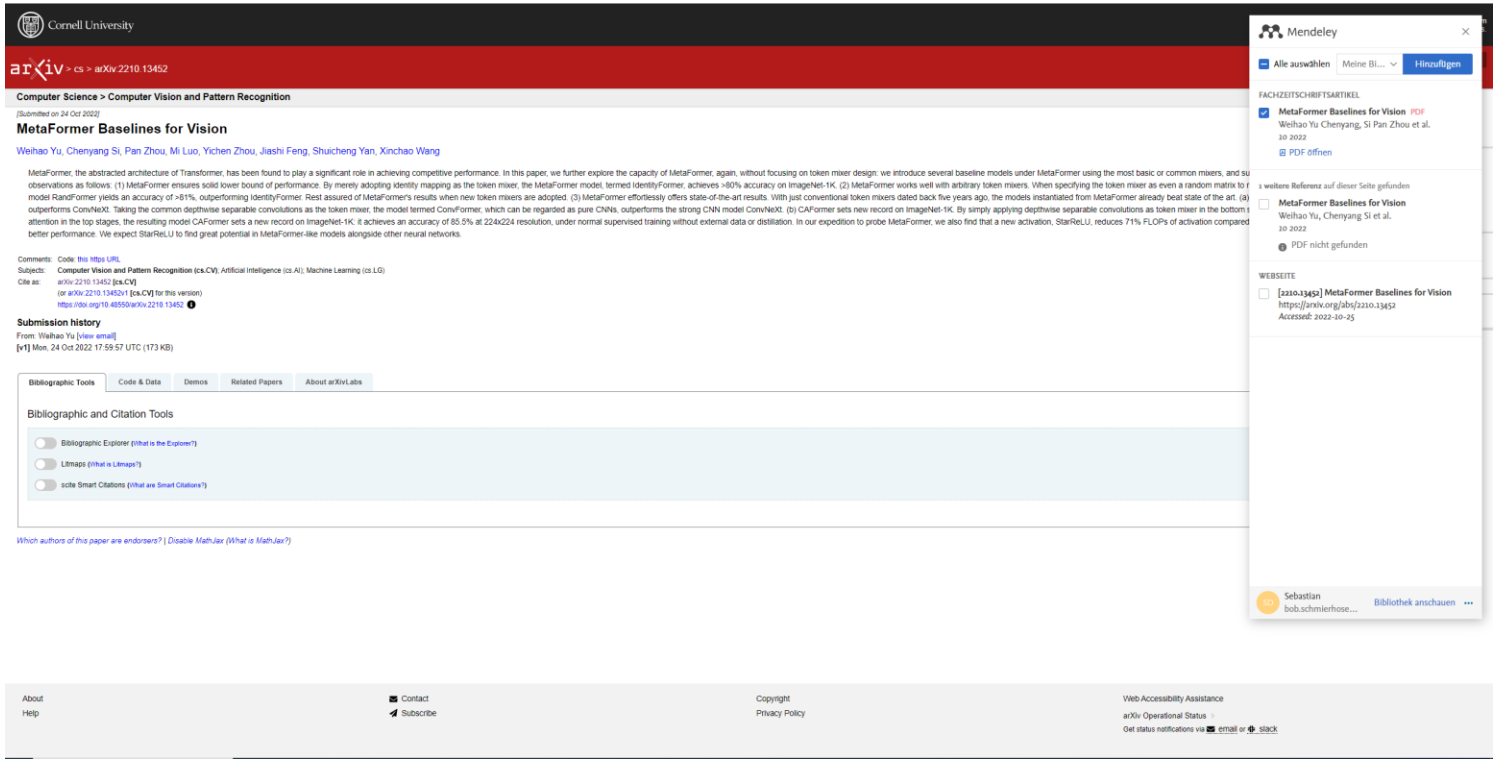
Label 3: NET. It is possible to identify such regions depicting the non-enhancing gross abnormality, by viewing the T2-weighted images. Some parts of the high-grade tumor do not enhance, but they are clearly distinguishable from the surrounding vasogenic edema on T2, as they have lower signal intensity and heterogeneous texture. Moreover, in low grade gliomas, this is the only category used for delineating the gross tumor.

Label 4: AT. This is a relatively easy definition, as it describes the enhancing regions within the gross tumor abnormality, but not the necrotic center. The threshold to exclude the necrotic center from the enhancing part should be set independently per subject. Note that vessels running in the neighboring regions and sulci are not included.

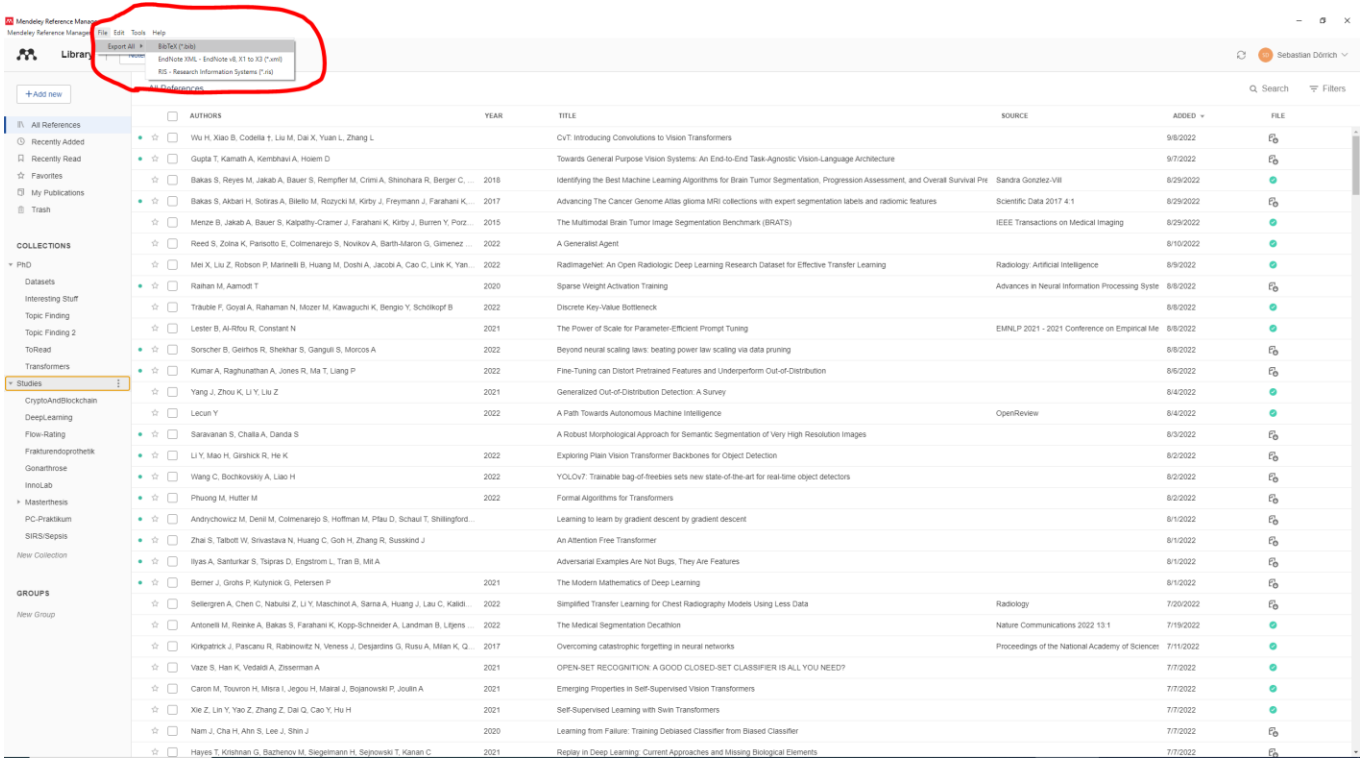
We cautiously note that the NET (i.e., "Label 3") can be overestimated by some annotators, and that oftentimes there is little evidence in the image data for this sub-region. Therefore, forcing the definition of this region could introduce an artifact, which could result in substantially different ground truth labels created from the annotators in different institutions. This case could potentially have implications in the ranking of the BraTS participants, i.e., a ranking bias towards the test cases ground truth annotator instead of ranking the actual algorithmic performance.

2.2.2 BraTS 2017-Present (Three tumor sub-regions) In order to address the aforementioned issue, in BraTS 2017 the NET label ("Label 3") has been eliminated and combined with NCR ("Label 1"). Furthermore, contralateral and periventricular regions of T2-FLAIR hyper-intensity were excluded from the ED region, unless they were contiguous with peritumoral ED, as these areas are generally considered to represent chronic microvascular changes, or age-associated demyelination, rather than tumor infiltration [7]. The rationale for this is that contralateral and periventricular white matter hyper-intensities regions might be considered pre-existing conditions, related to small vessel ischemic disease, especially in older patients.

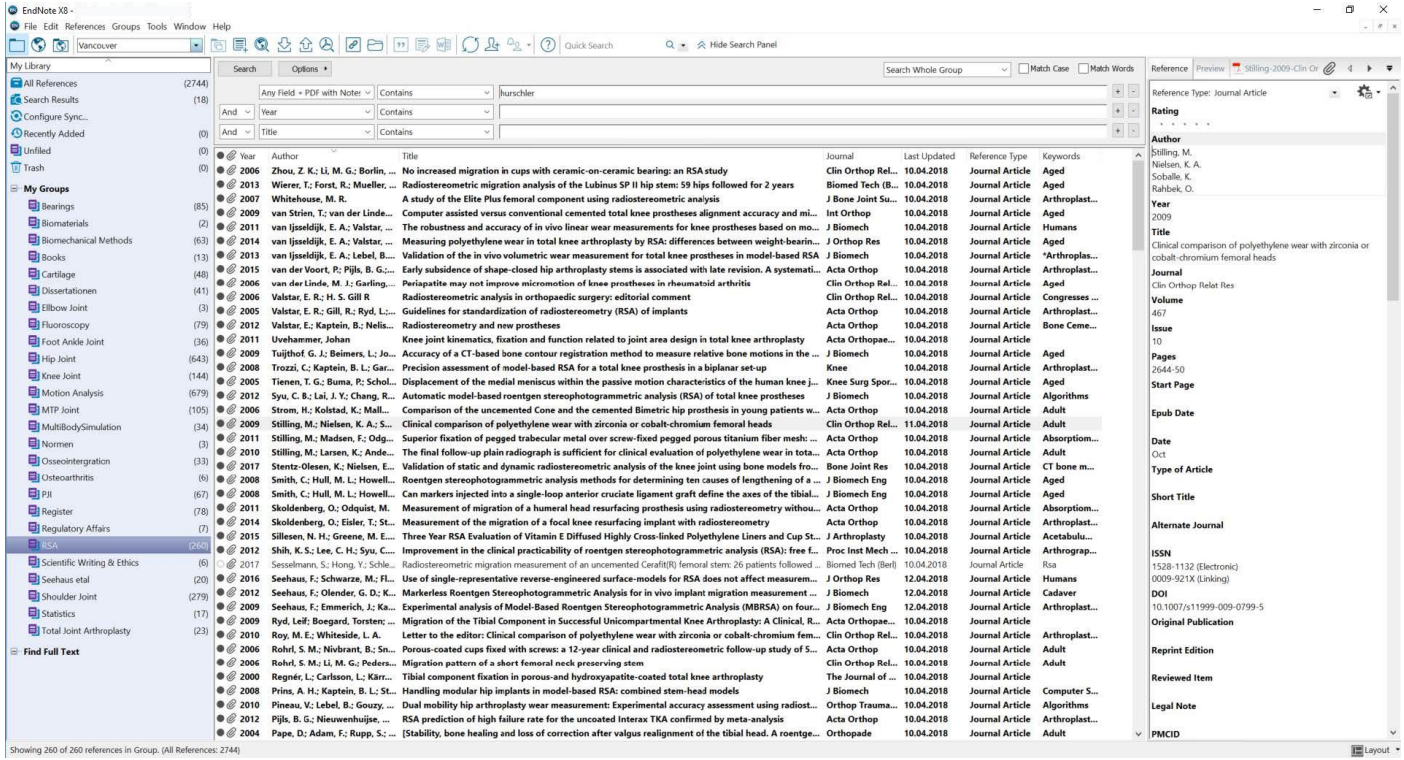
Mendeley Reference Manager – Web Importer



Mendeley Reference Manager – Bib Exporter



EndNote


EndNote X8 -
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Vancouver
Quick Search
Hide Search Panel
My Library
All References (2744)
Search Results (18)
Configure Sync...
Recently Added (0)
Unfiled (0)
Trash (0)
My Groups
Bearings (85)
Biomaterials (2)
Biomechanical Methods (63)
Books (13)
Cartilage (48)
Dissertationen (41)
Ellbow Joint (3)
Fluoroscopy (79)
Foot Ankle Joint (36)
Hip Joint (643)
Knee Joint (144)
Motion Analysis (679)
MTP Joint (105)
MultiBodySimulation (34)
Normen (3)
Osseointergration (33)
Osteoarthritis (6)
PJI (67)
Register (78)
Regulatory Affairs (7)
RSA (260)
Scientific Writing & Ethics (6)
Seehaus etal (20)
Shoulder Joint (279)
Statistics (17)
Total Joint Arthroplasty (23)
Find Full Text
Search
Options
Search Whole Group
Match Case
Match Words
Any Field + PDF with Notes
Contains
hurschler
And
Year
Title
Reference
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Stilling-2009-Clin Or
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Rating
Author
Stilling, M.
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Soballe, K.
Rahbek, O.
Year
2009
Title
Clinical comparison of polyethylene wear with zirconia or cobalt-chromium femoral heads
Journal
Clin Orthop Relat Res
Volume
467
Issue
10
Pages
2644-50
Start Page
Epub Date
Date
Oct
Type of Article
Short Title
Alternate Journal
ISSN
1528-1132 (Electronic)
0009-921X (Linking)
DOI
10.1007/s11999-009-0799-5
Original Publication
Reprint Edition
Reviewed Item
Legal Note
PMCID
Year Author Title Journal Last Updated Reference Type Keywords
2006 Zhou, Z. K.; Li, M. G.; Borlin, ... No increased migration in cups with ceramic-on-ceramic bearing: an RSA study Clin Orthop Rel... 10.04.2018 Journal Article Aged
2013 Wierer, T.; Forst, R.; Mueller, ... Radiostereometric migration analysis of the Lubinus SP II hip stem: 59 hips followed for 2 years Biomed Tech (B... 10.04.2018 Journal Article Aged
2007 Whitehouse, M. R. A study of the Elite Plus femoral component using radiostereometric analysis J Bone Joint Su... 10.04.2018 Journal Article Arthroplast...
2009 van Strien, T.; van der Linde... Computer assisted versus conventional cemented total knee prostheses alignment accuracy and mi... Int Orthop 10.04.2018 Journal Article Aged
2011 van Ijsseldijk, E. A.; Valstar, ... The robustness and accuracy of in vivo linear wear measurements for knee prostheses based on mo... J Biomech 10.04.2018 Journal Article Humans
2014 van Ijsseldijk, E. A.; Valstar, ... Measuring polyethylene wear in total knee arthroplasty by RSA: differences between weight-bearin... J Orthop Res 10.04.2018 Journal Article Aged
2013 van Ijsseldijk, E. A.; Lebel, B.... Validation of the in vivo volumetric wear measurement for total knee prostheses in model-based RSA J Biomech 10.04.2018 Journal Article *Arthroplas...
2015 van der Voort, P.; Pijls, B. G.;... Early subsidence of shape-closed hip arthroplasty stems is associated with late revision. A systemati... Acta Orthop 10.04.2018 Journal Article Arthroplast...
2006 van der Linde, M. J.; Garling,... Periapatite may not improve micromotion of knee prostheses in rheumatoid arthritis Clin Orthop Rel... 10.04.2018 Journal Article Aged
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2005 Valstar, E. R.; Gill, R.; Ryd, L.;... Guidelines for standardization of radiostereometry (RSA) of implants Acta Orthop 10.04.2018 Journal Article Arthroplast...
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2011 Uvehammer, Johan Knee joint kinematics, fixation and function related to joint area design in total knee arthroplasty Acta Orthopae... 10.04.2018 Journal Article
2009 Tuijthof G. J.; Beimers, L.; Jo... Accuracy of a CT-based bone contour registration method to measure relative bone motions in the ... J Biomech 10.04.2018 Journal Article Aged
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2005 Tienen, T. G.; Buma, P.; Schol... Displacement of the medial meniscus within the passive motion characteristics of the human knee j... Knee Surg Spor... 10.04.2018 Journal Article Aged
2012 Syu, C. B.; Lai, J. Y.; Chang, R... Automatic model-based roentgen stereophotogrammetric analysis (RSA) of total knee prostheses J Biomech 10.04.2018 Journal Article Algorithms
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2009 Stilling, M.; Nielsen, K. A.; S... Clinical comparison of polyethylene wear with zirconia or cobalt-chromium femoral heads Clin Orthop Rel... 11.04.2018 Journal Article Adult
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2008 Smith, C.; Hull, M. L.; Howell... Roentgen stereophotogrammetric analysis methods for determining ten causes of lengthening of a ... J Biomech Eng 10.04.2018 Journal Article Aged
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2011 Skoldenberg, O.; Odquist, M. Measurement of migration of a humeral head resurfacing prosthesis using radiostereometry withou... Acta Orthop 10.04.2018 Journal Article Absorptiom...
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2015 Sillesen, N. H.; Greene, M. E.... Three Year RSA Evaluation of Vitamin E Diffused Highly Cross-linked Polyethylene Liners and Cup St... J Arthroplasty 10.04.2018 Journal Article Acetabulu...
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2012 Seehaus, F.; Olender, G. D.; K... Markerless Roentgen Stereophotogrammetric Analysis for in vivo implant migration measurement ... J Biomech 12.04.2018 Journal Article Cadaver
2009 Seehaus, F.; Emmerich, J.; Ka... Experimental analysis of Model-Based Roentgen Stereophotogrammetric Analysis (MBRSA) on four... J Biomech Eng 12.04.2018 Journal Article Arthroplast...
2009 Ryd, Leif; Boegard, Torsten; ... Migration of the Tibial Component in Successful Unicompartmental Knee Arthroplasty: A Clinical, R... Acta Orthopae... 10.04.2018 Journal Article
2010 Roy, M. E.; Whiteside, L. A. Letter to the editor: Clinical comparison of polyethylene wear with zirconia or cobalt-chromium fem... Clin Orthop Rel... 10.04.2018 Journal Article Arthroplast...
2006 Rohrl, S. M.; Nivbrant, B.; Sn... Porous-coated cups fixed with screws: a 12-year clinical and radiostereometric follow-up study of 5... Acta Orthop 10.04.2018 Journal Article Adult
2006 Rohrl, S. M.; Li, M. G.; Peders... Migration pattern of a short femoral neck preserving stem Clin Orthop Rel... 10.04.2018 Journal Article Adult
2000 Regnér, L.; Carlsson, L.; Kärr... Tibial component fixation in porous-and hydroxyapatite-coated total knee arthroplasty The Journal of ... 10.04.2018 Journal Article
2008 Prins, A. H.; Kaptein, B. L.; St... Handling modular hip implants in model-based RSA: combined stem-head models J Biomech 10.04.2018 Journal Article Computer S...
2010 Pineau, V.; Lebel, B.; Gouzy, ... Dual mobility hip arthroplasty wear measurement: Experimental accuracy assessment using radiost... Orthop Trauma... 10.04.2018 Journal Article Algorithms
2012 Pijls, B. G.; Nieuwenhuijse, ... RSA prediction of high failure rate for the uncoated Interax TKA confirmed by meta-analysis Acta Orthop 10.04.2018 Journal Article Arthroplast...
2004 Pape, D.; Adam, F.; Rupp, S.; ... [Stability, bone healing and loss of correction after valgus realignment of the tibial head. A roentge... Orthopade 10.04.2018 Journal Article Adult
Showing 260 of 260 references in Group. (All References: 2744)
Layout

Other Reference Managers

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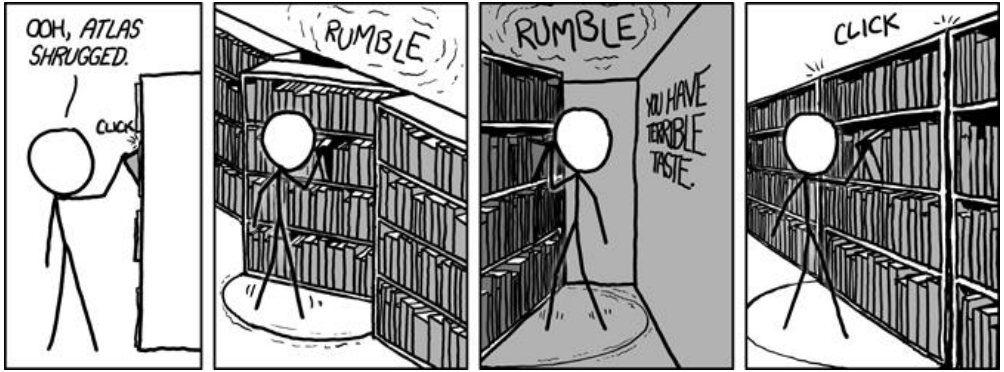
- Docear
- Citavi
- Wizfolio
- RefWorks
- Papers
- Sciwheel



Munroe, R. Xkcd. Bookshelf. https://xkcd.com/1049/

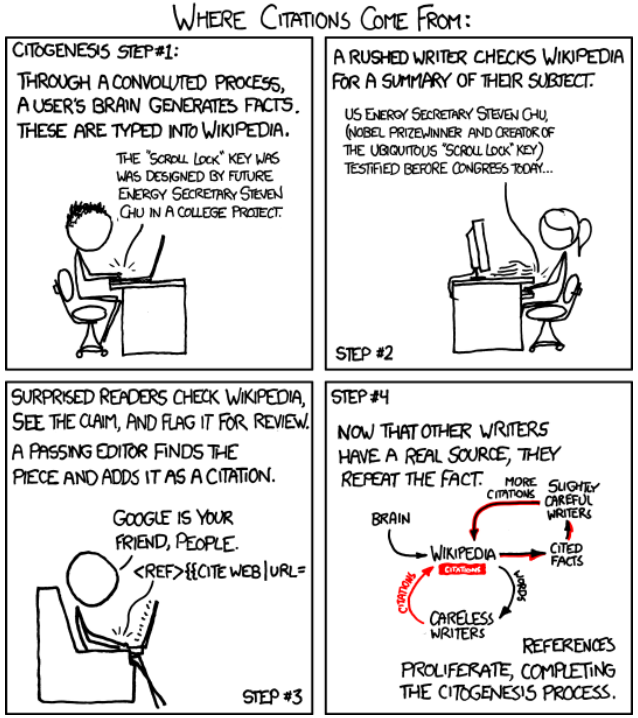
Citations

Guidelines

- Format dependent on the Conference/Journal/Chair
  - But always:
    - Consistent citation style
    - Consistent figure & table labeling
    - …
- Distinction
  - Journal/Reports/Theses
    - Citations within the text & bibliography
  - PowerPoint Presentations
    - Citations on the slides

What to cite?

- Textual content
  - Methods, Theories, Assumptions
  - Results
- Figures & Tables
- Medium
  - Journals
  - Conference Proceedings
  - (Text-)Books
  - **(**Internet sources**)**



Munroe, R. Xkcd. Citogenesis. https://xkcd.com/978/

How to cite?

“Reproduce intellectual property of another person in your own words”

Presentation

Evaluation Criteria

- Formalities
  - Timing
  - Appropriate references
- Key info on slides
- Avoidable technical problems

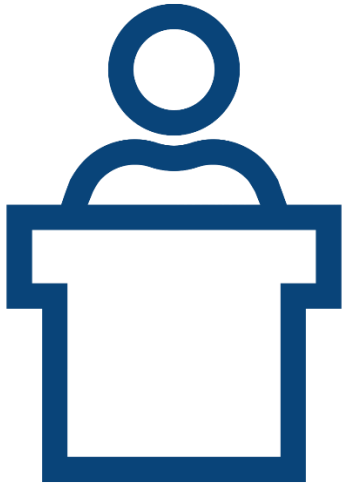
- Expertise/Content
  - Structure of Content
  - Appropriate Scope & Complexity
- Appropriate use of technical terms
- Educational for audience
- Knowledgeable Q&A

- Presentation Technique
  - Layout of Slides
  - Quality of Media
- Posture/eye contact w/ audience
- Clear communication/voice
- Respectful/non-defensive

Speech

- Just an act of performing?

- What is important?
  - Gestures
  - Speech
  - Sentence structure
  - Preliminary considerations
  - Elaborations



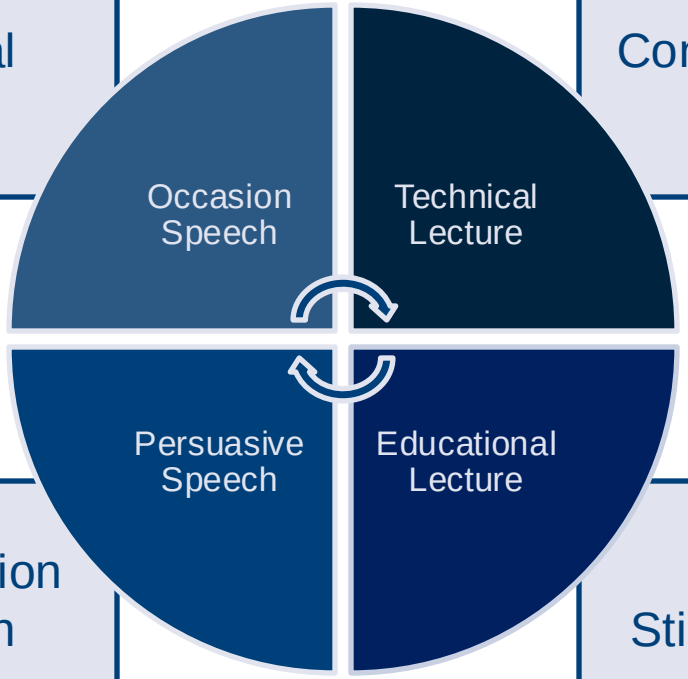
Speech – Occasions

Creation of a special mood

Emotional Factor
Person/occasion related

Convey knowledge and facts

From expert to expert(s)
(symmetrical ratio)



Problem processing
Proposed solution

Product demonstration
Formation of opinion

From expert to future expert(s)
(asymmetric relationship)

Get an overview
Stimulate own learning

Speech - Circumstances

- Audience:
  - What prior knowledge do they have?
  - Which interests do they have?
  - Which attitude do they have?

- Learning Objective
  - What is the learning objective?
  - What information do they need?
  - What should be remembered?

What do I need to achieve that ?

Speech – Stages Preparation

Choose Topic

- What do I talk about?
- Topic

Recognize Possibilities

- Facts
- Important Aspects

Disposing material

- Order
- Red thread

Structure Material Lingusistical

- Keywords
- Express content appropriatel y in terms of language

Memorize

- Know content / slides

Present

- Design talk
- Mimic
- Gestures
- Body language
- …

Speech – Outline

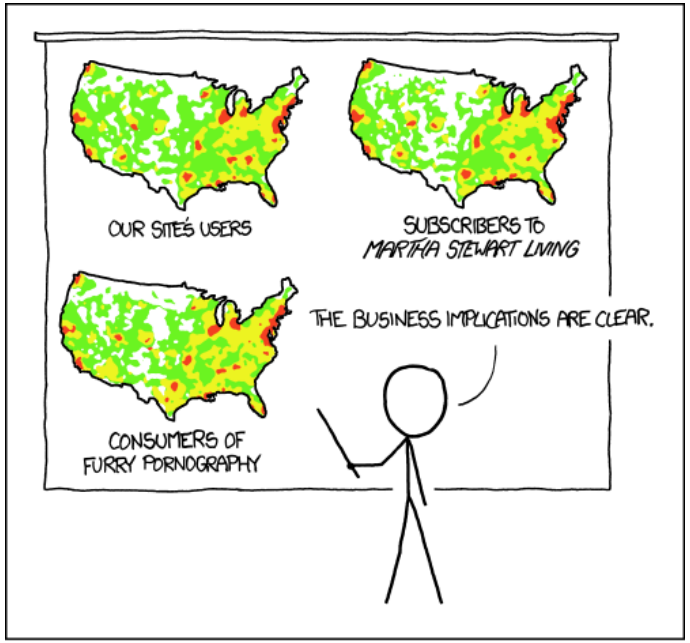
1. Introduction
   - Topic
   - Motivation
   - Background
2. Methods
3. Results
4. Discussion
5. Outlook

Speech – Presenting

- Eye contact with the audience
- Appropriate pace and volume
  - Variations can be appropriate
- Posture
  - Towards audience
  - Not just toward the screen
- Gestures
  - Should be supporting

Speech – Visualization

- Clarify complex relationships
- Increase attention
- Offer quick orientation
- Do not insert images arbitrarily
- No excessive animations



PET PEEVE #208:
GEOGRAPHIC PROFILE MAPS WHICH ARE BASICALLY JUST POPULATION MAPS

Munroe, R. Xkcd. Heatmap. https://xkcd.com/1138/

Citations

- Cite anything that is not your work (statements, figures, tables, …)
- Put the references directly on the respective slide (e.g. as footnotes)
- Collect all references at the end of the presentation as a bibliography
- Ensure that the references are readable but are not the main focus
- Avoid Wikipedia and online sources as much as possible
- Be consistent about your citation style

Citations – Citation suggestion for the slides

- Structure:
  - Author / Year / Journal (Abbrev. used by PubMed) / Volume / Issue / Page(s)
  - Author / Year / Book Title / Publisher / Page(s)

- Examples:
  - Budde et al. (2015) Int Orthop. 40(8):1607-1614
  - Valstar & Gill (2006) Clin Orthop Rel Res. 448:2
  - Maier et al. (2018) Medical Imaging Systems Springer: 253-254
  - Yuan & Jiang (2013) Diffuse Optical Tomography for Brain Imaging Springer: 105

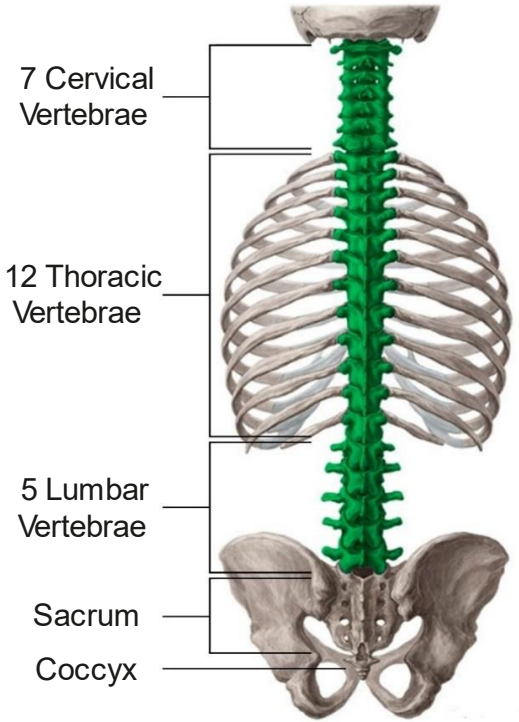
Citations – Example I

- Tasks of the human spine:
  - Connects the head to the pelvis
  - Protects the entire nerve supply
  - Allows for mobility

Farshad et al. (2021) Lehrbuch Orthopaedie: 7-8

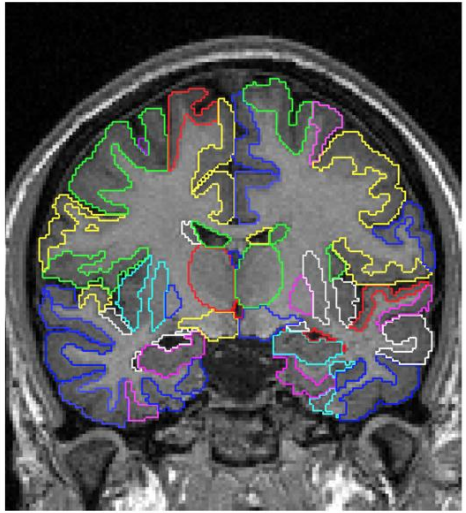
Schulter	Bizepssehnenruptur Bizepstendinitis Bizepssehneninstabilität AC-Arthritis Bursitis subdeltoidea et acromialis	Rotatorenläsion Tendinosis calcarea Bizepssehnendegeneration AC-Instabilität	Labrumläsionen Funktionelle Instabilität
Ellenbogen	Bursitis olecrani Gleitverhalten Radius-köpfchen	Freie Gelenkkörper Epicondylitis Bizepssehnenruptur	Instabilität Radiusköpfchenfraktur Suprakondyläre Frakturen
Hand	Funktionelle Sehnenunter-suchung Ganglion Tendovaginitis De Quervain	Tendinitis Palmare Platte Ringband	Skaphoidfraktur
Hüfte	Dysplasie Säugling Bursitis trochanterica	Bursitis iliopectinea	Labrum Kopfform Dynamische Untersuchung
Knie	Baker-Zyste Ruptur Patellasehne Ruptur Quadrizepssehne Meniskusganglion	Seitenbandläsion Patellaspitzensyndrom	Meniskusläsion Knorpelläsion
Fuß	Achillessehnenruptur Plantarfasziitis Ventrales Impingement	Außenband-, Innenbandriss Sehnenläsionen (Peronäal, Tibialis anterior et posterior, Flexor digitorum et hallucis)	
Gefäße	Screening Thrombose untere Extremität Qualitativer Nachweis arterieller Perfusion	Nachweis Thrombus	Beurteilung Thrombus

Tesch et al. (2008) Unfallchirurg 111(9):659

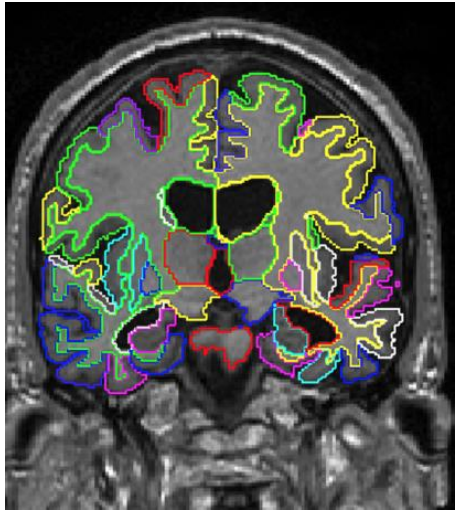


Frost et al. (2019) Materials (Basel) 12(2):253

Citations – Example II



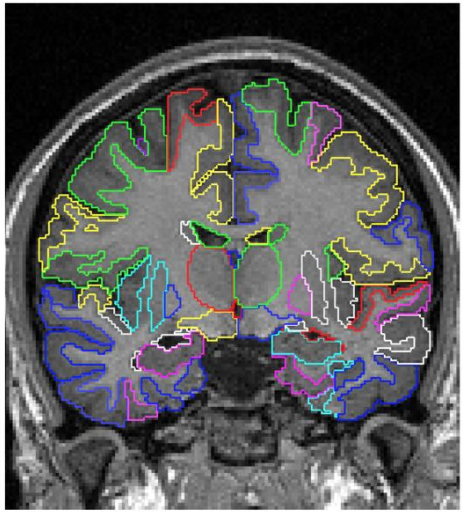
Healthy Subject



Subject with Alzheimer's

Ref.:
C. Ledig et al, "Structural brain imaging in Alzheimer's disease and mild cognitive impairment: biomarker analysis and shared morphometry database", Scientific Reports, 2018

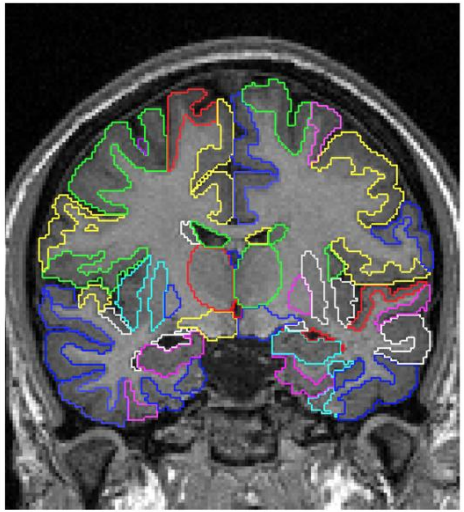
Citations – Examples III



Healthy Subject (Ledig 2018)

C. Ledig et al, “Structural brain imaging in Alzheimer’s disease and mild cognitive impairment: biomarker analysis and shared morphometry database”, Scientific Reports, 2018

Citations – Examples IV



Healthy Subject [1]

[1] C. Ledig et al, “Structural brain imaging in Alzheimer’s disease and mild cognitive impairment: biomarker analysis and shared morphometry database”, Scientific Reports, 2018

Citations – Examples V

Citations – Bibliography Example I

Simranjit Singh and Singara Singh Kasana. Efficient classification of the hyperspectral images using deep learning. Multimedia Tools and Applications, 77(20):27061–27074, 2018.

Satya P. Singh, Lipo Wang, Sukrit Gupta, Haveesh Goli, Parasuraman Padmanabhan, and Balázs Gulyás. 3D Deep Learning on Medical Images: A Review. Technical report, 2020.

Marina Sokolova and Guy Lapalme. A systematic analysis of performance measures for classification tasks. Information Processing and Management, 45:427–437, 2009.

Sabine Thürauf, Oliver Hornung, Mario Körner, Florian Vogt, Alois Knoll, and M. Ali Nasseri. Model-Based Calibration of a Robotic C-Arm System Using X-Ray Imaging. Journal of Medical Robotics Research, 2018

Citations – Bibliography Example II

[1] Simranjit Singh and Singara Singh Kasana. Efficient classification of the hyperspectral images using deep learning. Multimedia Tools and Applications, 77(20):27061–27074, 2018.

[2] Satya P. Singh, Lipo Wang, Sukrit Gupta, Haveesh Goli, Parasuraman Padmanabhan, and Balázs Gulyás. 3D Deep Learning on Medical Images: A Review. Technical report, 2020.

[3] Marina Sokolova and Guy Lapalme. A systematic analysis of performance measures for classification tasks. Information Processing and Management, 45:427–437, 2009.

[4] Sabine Thürauf, Oliver Hornung, Mario Körner, Florian Vogt, Alois Knoll, and M. Ali Nasseri. Model-Based Calibration of a Robotic C-Arm System Using X-Ray Imaging. Journal of Medical Robotics Research, 2018

Citations – Bibliography Example III

- [Sin18]
  Simranjit Singh and Singara Singh Kasana. Efficient classification of the hyperspectral images using deep learning. Multimedia Tools and Applications, 77(20):27061–27074, 2018.

- [Sin20]
  Satya P. Singh, Lipo Wang, Sukrit Gupta, Haveesh Goli, Parasuraman Padmanabhan, and Balázs Gulyás. 3D Deep Learning on Medical Images: A Review. Technical report, 2020.

- [Sok09]
  Marina Sokolova and Guy Lapalme. A systematic analysis of performance measures for classification tasks. Information Processing and Management, 45:427–437, 2009.

- [Thü18]
  Sabine Thürauf, Oliver Hornung, Mario Körner, Florian Vogt, Alois Knoll, and M. Ali Nasseri. Model-Based Calibration of a Robotic C-Arm System Using X-Ray Imaging. Journal of Medical Robotics Research, 2018

Citations – Bibliography Example IV

Tips & Tricks

- Create easy plots/diagrams/graphs on your own! Don't copy-paste!
- Re-create tables on your own! Don't copy-paste!
- Reduce tables to only the things you explain and avoid huge tables.
- Avoid notes if possible and try to speak freely
- A presenter can be useful for your posture
- It may be reasonable to reveal a slide little by little
- A rule of thumb: Each slide should take you about 1min to explain

Most Important!

- No whole sentences on the slides!
- Try to use nouns instead of verbs whenever possible!
- Keep the text to a minimum!
- Only use figures of appropriate quality and size!
- The slides are only for underlying and visualizing your arguments!
  The slides should never replace the talk!
  The audience should always focus on you, not the slides!
  Because they are interested in what you say, not what’s on the slides!

Report

Structure

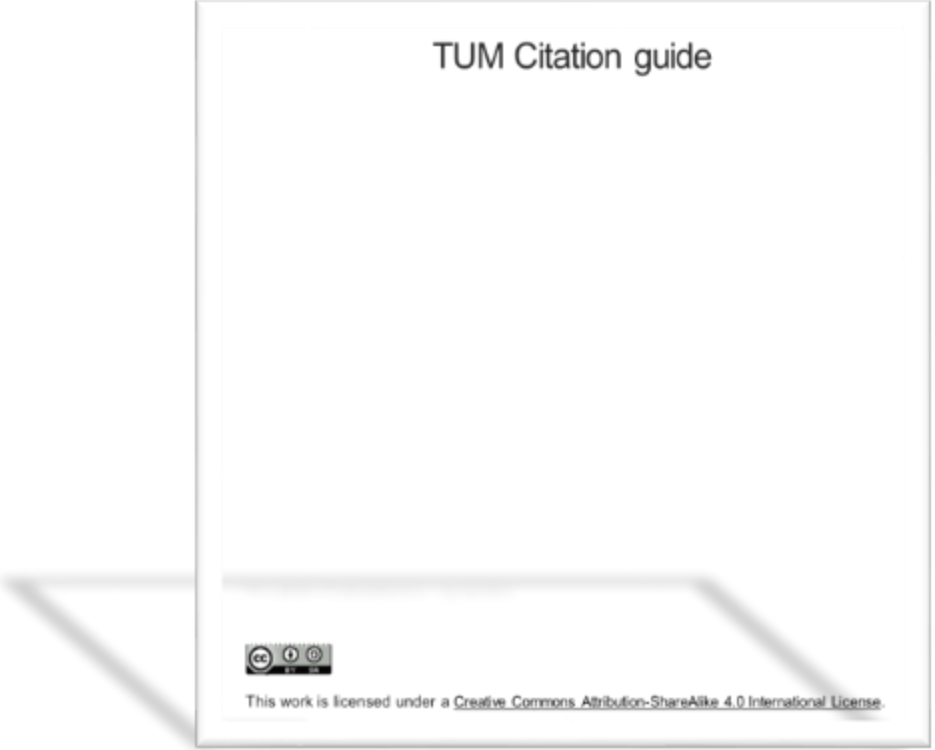
- Abstract (motivation, purpose, method, key findings, …)
- Introduction (motivation, background, related work, …)
- Methods (technical contribution, experimental setup, data, statistics, …)
- Results
- Discussion (positive findings, limitations, key findings, …)
- Conclusion (summary, outlook)
- References (bibliography)
- (Appendix (large tables and figures))

Evaluation Criteria – Report

- Formalities
  - Appropriate length
  - Appropriate references (inter)
  - Appropriate references (intra)
  - Key info present
- Content
  - Abstract & Introduction
  - Main Body
  - Discussion & Summary
  - Technical understanding
  - Educational & clear to follow
- Presentation
  - Logic/Structure
  - Appropriate use of tech. terms
  - Quality of layout and graphics
  - Readability/Language

Citations/References

- Citation Guide
  - From TU Munich
  - Very nice summary
  - Please read it
  - [Link]



Tips & Tricks II

- Be precise and objective
- Define the goal and key message of your report
- Use consistent terminology/symbols
- Introduce concepts, use established keywords
- Define abbreviations at first appearance
- No unnecessary complexity – Be educational
- Consider your audience

Tips & Tricks II

- Create easy plots/diagrams/graphs on your own!
- Ensure a proper quality of your images
- Rule of thumb for tables: No vertical lines!
- For algorithms use pseudocode
- Make sure your pdf is not unnecessary big (file size)
- Avoid unnecessary words, judgments, and tautologies (e.g. ‘very’)
- Be open for criticism

Castrol Player Index Rankings		
Rank	**Player**	**Player Rating**
1	Toni Kroos	9.79
2	Karim Benzema	9.74
3	Arjen Robben	9.70
4	Thiago Silva	9.66
5	Oscar	9.63
6	Stefan de Vrij	9.59
7	Thomas Müller	9.56
8	David Luiz	9.54
9	James Rodriguez	9.51
10	Neymar Jr.	9.48

Doerrich. (2018) Development of a Flow-Rating for Exercises based on Tracking and Event Data. Bachelor Thesis. FAU. Erlangen:4

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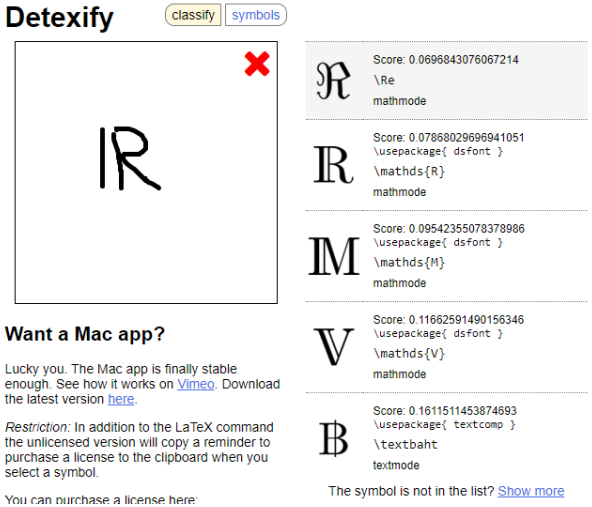
Tips & Tricks III

- Use a version control software (e.g. github)
- Search for unresolved references in your text [‘?’] before submission
- Ensure correct hyperlinks in your pdf (e.g. to chapters)
- Use the internet:
  - Google
  - Thesaurus.com
  - Dict.cc
  - Debugging of Latex problems
  - Stylistic advice

Most Important!

- Detexify
  - Draw symbols and get the latex command!
  - https://detexify.kirelabs.org/classify.html